Lie algebra cohomology

1 Chevalley–Eilenberg complex

Basic definitions. Let us start by reviewing the definition of the Chevalley–Eilenberg chain and cochain complex associated to any Lie algebra $\mathfrak{g}$.

Definition 1.1 [Chevalley–Eilenberg complex]**.** Let $(\mathfrak{g}, [-,-]_\mathfrak{g})$ be a Lie algebra and $(\mathbb{V}, \rho)$ be a $\mathfrak{g}$-module, with $\rho : \mathfrak{g} \to \mathrm{End}(\mathbb{V})$ defining the action of $\mathfrak{g}$ on the vector space $\mathbb{V}$. One can then define two complexes, namely

- The Chevalley–Eilenberg chain complex

$$\mathsf{CE}_\bullet(\mathfrak{g}, \mathbb{V}) := \bigoplus_{n\in\mathbb{N}} \mathsf{CE}_n(\mathfrak{g}, \mathbb{V})\,, \qquad \text{with} \qquad \mathsf{CE}_n(\mathfrak{g}, \mathbb{V}) := \mathbb{V} \otimes \wedge^n \mathfrak{g}\,, \tag{1.1}$$

  endowed with the (degree -1) boundary operator $\partial_\mathfrak{g} : \mathsf{CE}_\bullet(\mathfrak{g}, \mathbb{V}) \to \mathsf{CE}_{\bullet-1}(\mathfrak{g}, \mathbb{V})$ whose action on a chain of degree n reads

$$\begin{aligned} \partial_\mathfrak{g}(v \otimes x_1 \wedge \cdots \wedge x_n) \;&=\; \sum_{k=1}^{n} (-1)^{k+1}\, \rho(x_k)\, v \otimes x_1 \wedge \cdots \wedge \hat{x}_k \wedge \cdots \wedge x_n \\ &\quad + \sum_{1\leqslant i<j\leqslant n} (-1)^{i+j}\, v \otimes [x_i, x_i]_\mathfrak{g} \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge \hat{x}_j \wedge \cdots \wedge x_n\,, \end{aligned} \tag{1.2}$$

  where $v \in \mathbb{V}$ and $x_1, \dots, x_n \in \mathfrak{g}$;

- The Chevalley–Eilenberg cochain complex,

$$\mathsf{CE}^\bullet(\mathfrak{g}, \mathbb{V}) := \bigoplus_{n\in\mathbb{N}} \mathsf{CE}^n(\mathfrak{g}, \mathbb{V})\,, \qquad \text{with} \qquad \mathsf{CE}^n(\mathfrak{g}, \mathbb{V}) := \mathrm{Hom}(\wedge^n\mathfrak{g}, \mathbb{V}) \cong \wedge^n \mathfrak{g}^* \otimes \mathbb{V}\,. \tag{1.3}$$

  endowed with the (degree $+1$) differential $\delta_\mathfrak{g} : \mathsf{CE}^\bullet(\mathfrak{g}, \mathbb{V}) \to \mathsf{CE}^{\bullet+1}(\mathfrak{g}, \mathbb{V})$ whose action on a cochain of degree n reads

$$\begin{aligned} (\delta_\mathfrak{g}\Phi)(x_0, \dots, x_n) \;&=\; \sum_{k=1}^{n} (-1)^{k}\, \rho(x_k)\, \Phi(x_0, \dots, \hat{x}_k, \dots, x_n) \\ &\quad + \sum_{0\leqslant i<j\leqslant n} (-1)^{i+j}\, \Phi\big([x_i, x_j]_\mathfrak{g}, \dots, \hat{x}_i, \dots, \hat{x}_j, \dots, x_n\big)\,, \end{aligned} \tag{1.4}$$

  where $\Phi \in \mathrm{Hom}(\mathfrak{g}^{\wedge n}, \mathbb{V})$ and $x_0, \dots, x_n \in \mathfrak{g}$.

Note that $\partial_\mathfrak{g}$ and $\delta_\mathfrak{g}$ are dual to one another in the following sense. First, notice that the chain and cochain complexes of a representation and its dual (or contragredient) representation are dual as vector spaces, i.e.

$$\mathsf{CE}_\bullet(\mathfrak{g}, \mathbb{V})^* \cong \mathsf{CE}^\bullet(\mathfrak{g}, \mathbb{V}^*)\,. \tag{1.5}$$

As a consequence, the action of the boundary operator $\partial_\mathfrak{g}$ on a chain $c \in \mathsf{CE}_n(\mathfrak{g}, \mathbb{V})$ and is related to the action of the differential $\delta_\mathfrak{g}$ on a cochain chain $\phi \in \mathsf{CE}^{n-1}(\mathfrak{g}, \mathbb{V}^*)$ through

$$\langle \delta_\mathfrak{g}\phi, c\rangle = \langle \phi, \partial_\mathfrak{g} c\rangle\,, \tag{1.6}$$

where $\langle -,-\rangle$ denotes the pairing between dual vector spaces.

Choosing a basis $\{\mathsf{t}_a\}$ of $\mathfrak{g}$, in which the structure constants read $[\mathsf{t}_a, \mathsf{t}_b]_\mathfrak{g} = f_{ab}{}^c\, \mathsf{t}_c$, and a basis $\{\mathsf{t}_I\}$ of $\mathbb{V}$, in which the components of the representation read $\rho(\mathsf{t}_a)\,\mathsf{t}_I = \rho_a{}^J{}_I\, \mathsf{t}_J$, a n-cochain $\Phi \in \mathsf{CE}^n(\mathfrak{g}, \mathbb{V})$ takes the form

$$\Phi = \Phi^I_{a_1\dots a_n}\, \mathsf{t}^{a_1} \wedge \cdots \wedge \mathsf{t}^{a_n} \otimes \mathsf{t}_I\,, \tag{1.7}$$

where $\{\mathtt{t}^a\}$ is a basis of $\mathfrak{g}^*$, dual to the previously introduced one for $\mathfrak{g}$, and the cocycle condition $\delta_{\mathfrak{g}}\Phi = 0$ is given by

$$0 = \rho_{[a_0}{}^I{}_J\, \Phi^J_{a_1\dots a_n]} - \tfrac{1}{2}\, f_{[a_0 a_1}{}^b\, \Phi^I_{b|a_2\dots a_n]}\,. \tag{1.8}$$

A convenient way to represent these cochains and the Chevalley–Eilenberg differential, particularly when it comes to explicit computations, is to introduce Grassmann odd variables $\{\xi^a\}$, which stand in for the basis elements $\{\mathtt{t}^a\}$ and whose anticommuting nature account for the fact that we are interested in element of the exterior algebra $\wedge\mathfrak{g}^*$ tensored with a representation space $\mathbb{V}$. Consequently, cochains can be simply written as polynomials in ξ^a tensored with an element of $\mathbb{V}$, and the Chevalley–Eilenberg differential as

$$\delta_{\mathfrak{g}} = \xi^a \otimes \rho(\mathtt{t}_a) - \tfrac{1}{2}\, \xi^a \xi^b\, f_{ab}{}^c\, \tfrac{\partial}{\partial \xi^c} \otimes \mathbf{1}_{\mathbb{V}}\,, \tag{1.9}$$

where $\mathbf{1}_{\mathbb{V}}$ denotes the identity on the vector space $\mathbb{V}$.

The first few co/homology groups of the Chevalley–Eilenberg co/chain complex encode several characteristic of the Lie algebra $\mathfrak{g}$ and the representation $\mathbb{V}$. In degree 0, all chains $\mathsf{CE}_0(\mathfrak{g},\mathbb{V}) \equiv \mathbb{V}$ are closed, and the image of the boundary operator

$$\partial_{\mathfrak{g}}\mathsf{CE}_0(\mathfrak{g},\mathbb{V}) = \big\{\rho(x)v \mid x \in \mathfrak{g}\,,\ v \in \mathbb{V}\big\} \equiv \mathfrak{g}\mathbb{V}\,, \tag{1.10}$$

is the space of element which can be reached by the action of $\mathfrak{g}$ on $\mathbb{V}$, so that the homology group is simply the quotient

$$H_0(\mathfrak{g},\mathbb{V}) = \mathbb{V}/\mathfrak{g}\mathbb{V}\,, \tag{1.11}$$

which is the space of $\mathfrak{g}$-orbits in $\mathbb{V}$, also called the space of $\mathfrak{g}$-*coinvariant* of $\mathbb{V}$. The cohomology group

$$H^0(\mathfrak{g},\mathbb{V}) = \big\{v \in \mathbb{V} \mid \rho(x)v = 0\,,\ \forall x \in \mathfrak{g}\big\}\,, \tag{1.12}$$

correspond to the space of $\mathfrak{g}$-invariants of $\mathbb{V}$.

In degree 1, the cocycle condition for $\alpha : \mathfrak{g} \longrightarrow \mathbb{V}$ reads

$$\delta_{\mathfrak{g}}\alpha = 0 \qquad \Longleftrightarrow \qquad \alpha([x,y]_{\mathfrak{g}}) = \rho(x)\alpha(y) - \rho(y)\alpha(x)\,, \qquad \forall\, x,y \in \mathfrak{g}\,, \tag{1.13}$$

and means that α is a $\mathbb{V}$-valued $\mathfrak{g}$-derivation: thinking of the Lie bracket as a product, the above equation is the Leibniz identity for α, as it reads as the distribution of the action of α on both arguments of the Lie bracket, and the result is acted upon by the other argument via the representation of $\mathfrak{g}$ on $\mathbb{V}$. It may be useful to have a look at the case $\mathbb{V} = \mathfrak{g}$ to gain some intuition. Since $\rho(x) = [x,-]_{\mathfrak{g}}$ for the adjoint representation, the previous identity becomes

$$\alpha([x,y]_{\mathfrak{g}}) = [\alpha(x),y]_{\mathfrak{g}} + [x,\alpha(y)]_{\mathfrak{g}}\,, \tag{1.14}$$

which is easier to recognise as a Leibniz rule for the Lie bracket at first glance. A 1-cocycle is exact if it is of the form $\alpha(x) = \rho(x)v$ for some fixed element $v \in \mathbb{V}$. Such a map defines a particular type of derivation called *inner*, the space of which we denote by

$$\mathrm{Inn}(\mathfrak{g},\mathbb{V}) = \big\{\rho(-)v : \mathfrak{g} \longrightarrow \mathbb{V} \mid v \in \mathbb{V}\big\} \subset \mathrm{Der}(\mathfrak{g},\mathbb{V})\,, \tag{1.15}$$

where $\mathrm{Der}(\mathfrak{g},\mathbb{V})$ is the space of $\mathbb{V}$-valued $\mathfrak{g}$-derivations.

$$H^1(\mathfrak{g},\mathbb{V}) = \mathrm{Der}(\mathfrak{g},\mathbb{V})/\mathrm{Inn}(\mathfrak{g},\mathbb{V})\,, \tag{1.16}$$

i.e. the first Chevalley–Eilenberg cohomology group encodes $\mathbb{V}$-valued *outer* $\mathfrak{g}$-derivations.

In degree 2, the cocycle condition for a 2-cochain $\beta : \wedge^2\mathfrak{g} \longrightarrow \mathbb{V}$ reads

$$\begin{aligned}\delta_{\mathfrak{g}}\beta = 0 \qquad \Longleftrightarrow \qquad & \rho(x)\beta(y,z) - \rho(y)\beta(x,z) + \rho(z)\beta(x,y) \\ & - \beta([x,y]_{\mathfrak{g}},z) + \beta([x,z]_{\mathfrak{g}},y) - \beta([y,z]_{\mathfrak{g}},x) = 0\,,\end{aligned} \tag{1.17}$$

for any $x,y,z \in \mathfrak{g}$. In order to understand this condition, it is instructive to go back to the definition of the *semi-direct sum* of $\mathfrak{g}$ and $\mathbb{V}$, denoted by $\mathfrak{g} \in_\rho \mathbb{V}$. As a reminder, this simply the Lie algebra defined on the direct sum $\mathfrak{g} \oplus \mathbb{V}$ via the Lie bracket

$$[x \oplus v, y \oplus w]_{\mathfrak{g}\in_\rho\mathbb{V}} = [x,y]_{\mathfrak{g}} \oplus \rho(x)w - \rho(y)v\,, \qquad x,y \in \mathfrak{g}\,,\ v,w \in \mathbb{V}\,, \tag{1.18}$$

i.e. by the Lie bracket of $\mathfrak{g}$ and the antisymmetrisation of the action of $\mathfrak{g}$ and $\mathbb{V}$. Note in particular that $[v,w]_{\mathfrak{g}\Join_\rho\mathbb{V}}=0$ for $v,w\in\mathbb{V}$, so that $\mathbb{V}$ is an Abelian subalgebra. Now say that we want to try and deform this Lie bracket by adding a 2-cochain $\beta\in\wedge^2\mathfrak{g}^*\otimes\mathbb{V}$ to it,

$$[x\oplus v,y\oplus w]^\beta_{\mathfrak{g}\oplus\mathbb{V}}=[x,y]_{\mathfrak{g}}\oplus\rho(x)w-\rho(y)v+\beta(x,y)\,, \tag{1.19}$$

then the Jacobi identity is preserved when evaluated on at least one argument in $\mathbb{V}$, whereas if all entries belong to $\mathfrak{g}$, one finds

$$[x,[y,z]^\beta_{\mathfrak{g}\oplus\mathbb{V}}]^\beta_{\mathfrak{g}\oplus\mathbb{V}}-[[x,y]^\beta_{\mathfrak{g}\oplus\mathbb{V}},z]^\beta_{\mathfrak{g}\oplus\mathbb{V}}-[y,[x,z]^\beta_{\mathfrak{g}\oplus\mathbb{V}}]^\beta_{\mathfrak{g}\oplus\mathbb{V}}=(\delta_{\mathfrak{g}}\beta)(x,y,z)\,, \tag{1.20}$$

i.e. the Jacobi identity is verified for the bracket $[-,-]^\beta_{\mathfrak{g}\oplus\mathbb{V}}$ if and only if β is a Chevalley–Eilenberg 2-cocycle valued in $\mathbb{V}$. In other words, such cocycles encode *deformations* of the Lie bracket on $\mathfrak{g}\Join_\rho\mathbb{V}$ in the direction $\mathbb{V}$. For exact cocycle, i.e. $\beta(x,y)=\rho(x)\alpha(y)-\rho(y)\alpha(x)-\alpha([x,y]_{\mathfrak{g}})$, this kind of deformation is trivial in the sense that it leads to a Lie algebra structure isomorphic to the semi-direct sum $\mathfrak{g}\Join_\rho\mathbb{V}$ we started with. Indeed, the map $f\in\mathrm{End}(\mathfrak{g}\oplus\mathbb{V})$ defined by

$$f_\alpha(x\oplus v)=x\oplus v-\alpha(x)\,,\qquad x\in\mathfrak{g}\,,\ v\in\mathbb{V}\,, \tag{1.21}$$

for some 1-cochain $\alpha\in\mathfrak{g}^*\otimes\mathbb{V}$, is clearly an isomorphism of vector space, and verifies

$$[f_\alpha(x\oplus v),f_\alpha(y\oplus w)]^{\beta=\delta_{\mathfrak{g}}\alpha}_{\mathfrak{g}\oplus\mathbb{V}}=f_\alpha\big([x\oplus v,y\oplus w)]_{\mathfrak{g}\Join_\rho\mathbb{V}}\big)\,, \tag{1.22}$$

which tells us that it is an isomorphism of Lie algebras, as claimed. Looking for deformations of $\mathfrak{g}\Join_\rho\mathbb{V}$, it makes sense to discard these trivial deformations so that

$$H^2(\mathfrak{g},\mathbb{V})=\big\{\text{non-trivial deformations of }\mathfrak{g}\Join_\rho\mathbb{V}\big\}\,. \tag{1.23}$$

As a last comment, let us discuss the Chern–Simons 3-cocycle $\Phi_{\mathsf{cs}}:\wedge\mathfrak{g}^3\longrightarrow\mathbb{R}$, defined for any Lie algebra $\mathfrak{g}$ equipped with an ad-invariant symmetric form $\big\langle-,-\big\rangle$ via

$$\Phi_{\mathsf{cs}}(x,y,z)=\big\langle x,[y,z]_{\mathfrak{g}}\big\rangle\,, \tag{1.24}$$

for any $x,y,z\in\mathfrak{g}$. That it is closed under the Chevalley-Eilenberg (for the trivial representation) can be straightforwardly verified,

$$\begin{aligned}-\tfrac12\,\big(\delta_{\mathfrak{g}}\Phi_{\mathsf{cs}}\big)(w,x,y,z)&=\big\langle[w,x]_{\mathfrak{g}},[y,z]_{\mathfrak{g}}\big\rangle-\big\langle[w,y]_{\mathfrak{g}},[x,z]_{\mathfrak{g}}\big\rangle+\big\langle[w,z]_{\mathfrak{g}},[x,y]_{\mathfrak{g}}\big\rangle\\&=\big\langle w,[x,[y,z]_{\mathfrak{g}}]_{\mathfrak{g}}-[[x,y]_{\mathfrak{g}},z]_{\mathfrak{g}}-[y,[x,z]_{\mathfrak{g}}]_{\mathfrak{g}}\big\rangle=0\,,\end{aligned} \tag{1.25}$$

where in the first equality we merely regrouped pairs of identical terms due to the symmetry of the quadratic form (hence the factor $-\frac12$ on the left hand side in anticipation), and on the second line we used its ad-invariance to isolate the same argument in the first entry. The result vanishes, of course, because of the Jacobi identity, and therefore Φ_{cs} is indeed a 3-cocycle as claimed. Moreover, it is *not* exact since a coboundary would take the form

$$(\delta_{\mathfrak{g}}\varphi)(x,y,z)=-\varphi([x,y]_{\mathfrak{g}},z)+\varphi([x,z]_{\mathfrak{g}},y)-\varphi([y,z]_{\mathfrak{g}},x)\,, \tag{1.26}$$

for some $\varphi\in\wedge^2\mathfrak{g}^*$, and hence cannot reproduce Φ_{cs}. This implies that the Chern–Simons cocycle defines a cohomology class $[\Phi_{\mathsf{cs}}]\in H^3(\mathfrak{g},\mathbb{R})$, and the previous considerations shows that such a class exists for any Lie algebra $\mathfrak{g}$ equipped with a symmetric *ad-invariant* bilinear form.

Curvature from CE algebra. First, let us have a look at what it means for A to be a morphism of dgca. Recall that the Chevalley–Eilenberg differential $\delta_{\mathfrak{g}}$ is defined by its action of a basis $\{\mathsf{t}^a\}$ of $\mathfrak{g}^*$ as

$$\delta_{\mathfrak{g}}\mathsf{t}^a=-\tfrac12\,f_{bc}{}^a\,\mathsf{t}^b\wedge\mathsf{t}^c\,, \tag{1.27}$$

and extended to $\wedge\mathfrak{g}^*$ as a derivation. To avoid confusion, let us denote the $\mathfrak{g}$-valued 1-form A seen as a map from $\mathfrak{g}^*$ to Ω^1_Σ as

$$a:\mathfrak{g}^*\longrightarrow\Omega^1_\Sigma\,, \tag{1.28}$$

so that to a basis element t^a of $\mathfrak{g}^*$, it assigns the component of the gauge field along the dual generator t_a of $\mathfrak{g}$, i.e.

$$\Omega^1_\Sigma\otimes\mathfrak{g}\ni A=A^a\,\mathsf{t}_a\qquad\text{with}\qquad A^a:=a(\mathsf{t}^a)\in\Omega^1_\Sigma\,, \tag{1.29}$$

and the fact that a seen as a linear map between $\wedge\mathfrak{g}^*$ and Ω^1_Σ preserves the commutative products means

$$a(\mathsf{t}^a \wedge \mathsf{t}^b) = a(\mathsf{t}^a) \wedge a(\mathsf{t}^b) = A^a \wedge A^b \,. \tag{1.30}$$

Now we can write down what the requirement that A be a chain map, that is

$$a \circ \delta_{\mathfrak{g}} = \mathrm{d}_\Sigma \circ a \,, \tag{1.31}$$

entails. First, applying a to the image of t^a under the Chevalley–Eilenberg differential yields

$$a(\delta_{\mathfrak{g}} \mathsf{t}^a) = -\tfrac{1}{2}\, a(f_{bc}{}^a\, \mathsf{t}^b \wedge \mathsf{t}^c) = -\tfrac{1}{2}\, f_{bc}{}^a\, A^b \wedge A^c \tag{1.32}$$

and second, acting with the de Rham differential on a applied to t^a simply gives

$$\mathrm{d}_\Sigma\, a(\mathsf{t}^a) = \mathrm{d}_\Sigma A^a \,, \tag{1.33}$$

so that, putting together these equations we find

$$a \circ \delta_{\mathfrak{g}} - \mathrm{d}_\Sigma \circ a = 0 \qquad \Longleftrightarrow \qquad \mathrm{d}_\Sigma A + \tfrac{1}{2}\,[A, A]_{\mathfrak{g}} = 0 \,, \tag{1.34}$$

where we have switch back to seeing A as a $\mathfrak{g}$-valued 1-form on Σ in the second equation. In plain words, the gauge field A is a morphism of dgca between the Chevalley–Eilenberg algebra $\wedge\mathfrak{g}^*$ and the algebra of differential forms Ω_Σ if and only if its curvature vanishes.

2 Weil algebra

Let us start with the definition of the Weil algebra.

Definition 2.1 [Weil algebra]**.** Let $\mathfrak{g}$ be a Lie algebra. Its Weil algebra $\big(\mathrm{W}(\mathfrak{g}), \delta_W\big)$ is the differential graded-commutative and associative algebra (dgca), with underlying $\mathbb{N}$-graded vector space

$$\mathrm{W}(\mathfrak{g}) := \bigoplus_{n\in\mathbb{N}} \mathrm{W}_n(\mathfrak{g}) \,, \qquad \text{with} \qquad \mathrm{W}_n(\mathfrak{g}) := \bigoplus_{n=k+2l} \wedge^k\mathfrak{g}^* \otimes S^l(\mathfrak{g}^*) \,, \tag{2.1}$$

i.e. elements of $\wedge^1\mathfrak{g}^*$ are of degree 1 and those of $S^1(\mathfrak{g}^*)$ are of degree 2, together with the differential

$$\delta_W := \delta_K + \delta_{\mathfrak{g}} \,, \tag{2.2}$$

where

$$\delta_K : \wedge^p\mathfrak{g}^* \otimes S^q(\mathfrak{g}^*) \longrightarrow \wedge^{p-1}\mathfrak{g}^* \otimes S^{q+1}(\mathfrak{g}^*) \,, \tag{2.3}$$

is the extension of the suspension map

$$s : \wedge^1\mathfrak{g}^* \longrightarrow S^1(\mathfrak{g}^*) \,, \qquad s : S^1(\mathfrak{g}^*) \longrightarrow 0 \,, \tag{2.4}$$

as a derivation to $\mathrm{W}(\mathfrak{g})$ and called the Koszul differential, and

$$\delta_{\mathfrak{g}} : \wedge^p\mathfrak{g}^* \otimes S^q(\mathfrak{g}^*) \longrightarrow \wedge^{p+1}\mathfrak{g}^* \otimes S^q(\mathfrak{g}^*) \,, \tag{2.5}$$

is the Chevalley–Eilenberg differential with values in the module $S(\mathfrak{g}^*)$.

As for the Chevalley–Eilenberg algebra, we can characterise the differential δ_W by its action on generators. We will use the notation $\{\mathsf{t}^a\}$ for the basis of $\mathfrak{g}^*$ generating $\wedge\mathfrak{g}^*$, i.e. elements of degree 1, and $\{\mathsf{s}^a\}$ for the (same) basis of $\mathfrak{g}^*$ generating $S(\mathfrak{g}^*)$, i.e. elements of degree 2. With these notations, and degrees assigned, the Weil algebra is simply the algebra of polynomials in these generators,

$$\mathrm{W}(\mathfrak{g}) \cong \mathbb{R}[\mathsf{t}^a, \mathsf{s}^a] \,, \tag{2.6}$$

in the graded sense (meaning, keeping in mind that $\deg(\mathsf{t}^a) = 1$ and $\deg(\mathsf{s}^a) = 2$, and that they commute was prescribed by this grading). The action of the differential δ_W on these generators then reads

$$\delta_W \mathsf{t}^a = \mathsf{s}^a - \tfrac{1}{2}\, f_{bc}{}^a\, \mathsf{t}^b \wedge \mathsf{t}^c \,, \qquad \delta_W \mathsf{s}^a = -f_{bc}{}^a\, \mathsf{t}^b \otimes \mathsf{s}^a \,, \tag{2.7}$$

and its action is extended to the whole Weil algebra by derivation.

A basis-independent expression of the Koszul differential can be given as follows: its action on a cochain $\Phi \in \wedge^m \mathfrak{g}^* \otimes S\mathfrak{g}^*$ reads,

$$(\delta_K \Phi)(x_0, \dots, x_m|-) = \sum_{k=0}^{m} (-1)^m \, \Phi(x_0, \dots, \hat{x}_k, \dots, x_m|s(x_k), -) \,, \tag{2.8}$$

where the hatted arguments are omitted, and the dash stands in for entries in $S^1\mathfrak{g}^*$.

Similarly to the Chevalley–Eilenberg complexes discussed previously, one can realise the Weil algebra using Grassmann odd variables $\{\xi^a\}$ for the generators $\{\mathsf{t}^a\}$ of $\wedge\mathfrak{g}^*$, and Grassmann even variables σ^a for the generators $\{\mathsf{s}^a\}$ of $S(\mathfrak{g}^*)$. In terms of these variables, the Weil differential reads

$$\delta_W = \left(\sigma^a - \tfrac{1}{2} \, f_{bc}{}^a \, \xi^b \xi^c\right) \tfrac{\partial}{\partial \xi^a} + \xi^a \, f_{ab}{}^c \, \sigma^b \, \tfrac{\partial}{\partial \sigma^c} \,, \tag{2.9}$$

and can be interpreted as a cohomological vector field on the graded manifold $\mathfrak{g}[1] \oplus \mathfrak{g}[2]$.

Lemma 2.1. *The Weil algebra is acyclic, i.e. its cohomology is concentrated in degree* 0*.*

Proof. First, let us define the linear map $f : \mathrm{W}(\mathfrak{g}) \longrightarrow \mathrm{W}(\mathfrak{g})$ by

$$f(\mathsf{t}^a) = \mathsf{t}^a \,, \qquad f(\mathsf{s}^a) = \mathsf{s}^a + \delta_{\mathfrak{g}} \mathsf{t}^a \,, \tag{2.10}$$

and extending to obtain a morphism of graded commutative algebras. This clearly defines an isomorphism of graded commutative algebras, since

$$\mathrm{W}(\mathfrak{g}) \cong \mathbb{R}[\mathsf{t}^a, \mathsf{s}^a] \cong \mathbb{R}[f(\mathsf{t}^a), f(\mathsf{s}^a)] \,. \tag{2.11}$$

On top of that, a direct computation leads to

$$\delta_W f(\mathsf{t}^a) = f(\mathsf{s}^a) \,, \qquad \delta_W f(\mathsf{s}^a) = 0 \,, \tag{2.12}$$

which means that the Weil algebra is generated by trivial pairs, and hence has trivial cohomology. ∎

The Weil algebra provides us with a way of encoding connections on principal bundles and their curvatures. Indeed, thinking of

$$\begin{cases} \mathsf{t}^a & \longleftrightarrow & \text{gauge potential } A \,, \\ \mathsf{s}^a & \longleftrightarrow & \text{curvature } F \,, \\ \delta_W & \longleftrightarrow & \text{de Rham differential } \mathrm{d} \,, \end{cases} \tag{2.13}$$

the previous equations defining the action of δ_W on the generators of $\mathrm{W}(\mathfrak{g})$ can be rea as the definition of the curvature of a G-principal connection, and its Bianchi identity. Let us make this more precise.

Lemma 2.2. *Morphisms of dgca $\varphi : \mathrm{W}(\mathfrak{g}) \longrightarrow \Omega^1_\Sigma$ are in bijection with pairs (A, F) made out of a $\mathfrak{g}$-valued* 1*-form and its curvature* 2*-form.*

Proof. A morphism of dgca $\varphi : \mathrm{W}(\mathfrak{g}) \longrightarrow \Omega^1_\Sigma$ should preserve the commutative product on both sides, and hence is completely determined by its values on the generators of $\mathrm{W}(\mathfrak{g})$, that is

$$A^a := \varphi(\mathsf{t}^a) \,, \qquad F^a := \varphi(\mathsf{s}^a) \,, \tag{2.14}$$

or in a basis independent form

$$A \equiv A^a \, \mathsf{t}_a = \varphi|_{\wedge^1 \mathfrak{g}^*} \in \Omega^1_\Sigma \otimes \mathfrak{g} \,, \qquad F \equiv F^a \, \mathsf{s}_a = \varphi|_{S^1(\mathfrak{g}^*)} \in \Omega^2_\Sigma \otimes \mathfrak{g} \,. \tag{2.15}$$

On top of that, it should intertwine the differentials, meaning

$$\mathrm{d}_\Sigma \, \varphi = \varphi \, \delta_W \tag{2.16}$$

which, upon evaluation on $\wedge^1\mathfrak{g}^*$ and $S^1(\mathfrak{g}^*)$ yields

$$\begin{cases} F = \mathrm{d}_\Sigma A + \frac{1}{2} \, [A, A]_{\mathfrak{g}} \,, \\ 0 = \mathrm{d}_\Sigma F + [A, F]_{\mathfrak{g}} \,. \end{cases} \tag{2.17}$$

Conversely, it is clear that any such pair (A, F) defines a dgca morphism. ∎

With this problem solved, we can turn our attention to homotopies between dgca morphisms, i.e. dgca morphisms

$$H : \mathrm{W}(\mathfrak{g}) \longrightarrow \Omega_\Sigma \otimes \mathbb{R}[t, \mathrm{d}t]\,, \tag{2.18}$$

with appropriate boundary conditions (meaning the values of H at $t = 0$ and $t = 1$ corresponds to two gauge field, say A and A'). Since H preserves the commutative product on both sides, it is also completely determined by its values on generators of degree 1, in $\wedge^1\mathfrak{g}^*$ and 2, in $S^1(\mathfrak{g}^*)$, i.e.

$$H_t|_{\wedge^1\mathfrak{g}^*} = A_t + \mathrm{d}t\,\xi_t\,, \qquad H_t|_{S^1(\mathfrak{g}^*)} = F_t + \mathrm{d}t\,\Xi_t\,, \tag{2.19}$$

where, on Σ, the component ξ_t is a 0-form, A_t and Ξ_t are 1-forms and F_t is a 2-form, and where we have introduced a t subscript on all fields to keep in mind that there components depend on t. As suggested by the notation, we will see that ξ_t plays the role of a gauge parameter, however we will have no use for the component Ξ_t, and therefore we will only consider homotopies for which $\Xi_t = 0$. This particular type of homotopies are sometimes called *partially flat* (see e.g. [1, Sec. 2.1] for a closely related discussion). Now let us have a look at the condition that H_t intertwines the differentials in $W(\mathfrak{g})$ and in Ω_Σ: a direct computation yields

$$H_t\,\delta_W|_{\wedge^1\mathfrak{g}^*} = F_t - \tfrac{1}{2}\,[A_t, A_t]_\mathfrak{g} + \mathrm{d}t \wedge [A_t, \xi_t]_\mathfrak{g}\,, \tag{2.20a}$$

and

$$\mathrm{d}_{\Sigma\times[0,1]}\,H_t|_{\wedge^1\mathfrak{g}^*} = \mathrm{d}_\Sigma A_t + \mathrm{d}t\,\partial_t A_t - \mathrm{d}t \wedge \mathrm{d}_\Sigma\xi_t\,, \tag{2.20b}$$

for the evaluation on degree 1 generators of $\mathrm{W}(\mathfrak{g})$, and

$$H_t\,\delta_W|_{S^1(\mathfrak{g}^*)} = -[A_t, F_t]_\mathfrak{g} - \mathrm{d}t \wedge [\xi_t, F_t]_\mathfrak{g}\,, \qquad \mathrm{d}_{\Sigma\times[0,1]}\,H_t|_{S^1(\mathfrak{g}^*)} = \mathrm{d}_\Sigma F_t + \mathrm{d}t\,\partial_t F_t\,, \tag{2.20c}$$

on degree 2 generators, which leads to four conditions

$$H_t\,\delta_W - \mathrm{d}_\Sigma\,H_t = 0 \qquad \Longleftrightarrow \qquad \begin{cases} F_t = \mathrm{d}_\Sigma A_t + \frac{1}{2}\,[A_t, A_t]_\mathfrak{g}\,, \\ 0 = \mathrm{d}_\Sigma F_t + [A_t, F_t]_\mathfrak{g}\,, \\ \frac{\mathrm{d}A_t}{\mathrm{d}t} = \mathrm{d}_\Sigma\xi_t + [A_t, \xi_t]_\mathfrak{g}\,, \\ \frac{\mathrm{d}F_t}{\mathrm{d}t} = [F_t, \xi_t]_\mathfrak{g}\,. \end{cases} \tag{2.21}$$

The first and second equations are expected: they merely signifies that F_t is the curvature of A_t and obey the corresponding Bianchi identity, for any values of t. We have seen previously that this amount to saying that the pair (F_t, A_t) defines a morphism of dgca for all t. The third equation is what we were after: it gives us a way of defining what is an infinitesimal gauge symmetry for A. Indeed, since A_t defines a one-parameter family of gauge fields with $A_0 = A$, an infinitesimal gauge transformation of the latter can be taken as the first order in t of A_t, i.e.

$$\delta_\xi A := \tfrac{\mathrm{d}A_t}{\mathrm{d}t}|_{t=0} = \mathrm{d}_\Sigma\xi + [A, \xi]_\mathfrak{g}\,, \qquad \text{with} \qquad \xi := \xi_0\,, \tag{2.22}$$

as expected. Similarly, evaluating the fourth equation at $t = 0$ shows that the curvature $F \equiv F_0$ of A transforms in the adjoint representation of $\mathfrak{g}$ under a gauge transformation.

References

[1] H. Kim and C. Saemann, "Adjusted parallel transport for higher gauge theories," *J. Phys. A* **53** (2020) no. 44, 445206, arXiv:1911.06390 [hep-th].